

Quantitative evaluation of volatility-adaptive trend-following models in cryptocurrency markets

Ioannis Karassavidis, Athens College, ikarassavidis@athenscollege.edu.gr

Abstract

Cryptocurrency markets exhibit extreme volatility, heavy-tailed return distributions, and structurally persistent inefficiencies arising from fragmented liquidity and heterogeneous market participants. Despite extensive research on momentum and volatility-managed portfolios in traditional asset classes, empirical evidence for volatility-scaled trend-following in major cryptocurrency markets remains limited. In addition to posing a challenge to conventional portfolio theory, these features offer a special testing ground for momentum-based trading systems.

This paper presents a trend-following framework that adjusts to volatility in high-frequency cryptocurrency markets, focusing on Bitcoin (BTC) and Ethereum (ETH). The proposed structure uses multi-horizon moving averages to detect trends. It uses a multi-timeframe Relative Strength Index (RSI) filter to confirm momentum. Risk management is conducted through volatility-sensitive ATR scaling. The exits are generated based on the combination of ATR trailing stop with EMA slope reversal. Overall, the combination of all elements provides an exit method that strikes a balance between persistence and risk management.

Parameters are optimized via a large-scale Latin Hypercube Sampling (LHS) procedure featuring 5,000,000 samples across 5 configurations, evaluated under an objective function maximizing a weighted combination of Sharpe ratio, compound annual growth rate (CAGR), and maximum drawdown. From this high-dimensional search the procedure identifies a final configuration that performs consistently across both assets. When re-evaluated using the original backtest engine on 8-hour data from 2020–2025, the BTC strategy attains a Sharpe ratio of approximately ≈ 1.54 with a CAGR of $\approx 20\%$ and a maximum peak-to-trough drawdown of 12.5%. The ETH strategy achieves an even higher Sharpe ratio of ≈ 1.80 with a CAGR of $\approx 24.3\%$ and a maximum drawdown of $\approx 10.0\%$. Averaged across both assets, the final system delivers a Sharpe ratio of ≈ 1.67 , an average annual CAGR of $\approx 22.1\%$ and an average maximum drawdown of roughly ≈ 11.3 . The equity curve showcases the capital growing from \$10,000 to around \$30–32,000 when invested in Bitcoin, and \$36–38,000 when invested in Ethereum, over a period of five years. Notably, the strategy avoids major collapses, even during times of high market stress. These performance metrics incorporate transaction costs, which are based on the fee schedules of Binance Futures, this provides a more realistic estimate of how the investments would perform in real-life scenarios. The average annual CAGR across both assets comes out to 22.14%

Furthermore, cross-correlation analysis reveals that BTC and ETH share a time-varying but persistently high correlation structure ($\rho \approx 0.8\text{--}0.9$), suggesting common volatility regimes. This validates the use of joint parameter optimization across correlated assets as a robustness-enhancing constraint.

The results demonstrate that volatility-adjusted, momentum-confirmed trend-following systems can generate statistically and economically significant excess returns in cryptocurrency markets, provided that their optimization framework is adaptive, cross-asset validated, and sensitive to dynamic volatility states.

Keywords

cryptocurrency markets; trend-following; volatility scaling; Bitcoin; Ethereum; risk management; moving averages; ATR; Sharpe ratio.

Abbreviations

ATR, average true range; BTC, Bitcoin; ETH, Ethereum; EMA, exponential moving average; LHS, Latin hypercube sampling; MA, moving average; PnL, profit and loss; RSI, relative strength index; SMA, simple moving average.

1. Introduction

1.1 Background and motivation

Since the launch of Bitcoin futures in 2017 and the subsequent proliferation of exchange-traded cryptocurrency derivatives, the markets for cryptocurrencies have started drawing more interest from quantitative investors and researchers. Relative to traditional asset classes, major cryptocurrencies (e.g., BTC and ETH) appear to possess high levels of volatility, frequent trend reversals and moments of extreme liquidity and leverage. At the same time, there is evidence of medium-term momentum and trend-following effects in crypto spot and futures markets, especially in the wake of significant shifts in market direction and deleveraging.

These features make cryptocurrency markets an interesting place to test systematic trend-following strategies. The standard approach to trend-following strategies in futures involves MA crossovers, breakouts or momentum filters applied to time series, possibly in combination with volatility-adjusted sizing and fairly large stop-losses. In relation to the cryptocurrency market, the problem is twofold -elevated and less stable volatility, and fast-changing nature of the market due to the emergence of perpetual swaps, the mechanism of the funding rate, and sudden changes triggered by external events. A effective crypto trend-following strategy must be responsive enough to catch large directional movements, but adaptive enough to survive prolonged consolidation phases.

In that regard, this paper attempts to explore the applicability of the trend-following algorithms (commonly used in commodities and forex markets) to cryptocurrency markets with appropriate adjustment for volatility to evaluate whether it is possible to achieve strong risk-adjusted returns in the cryptocurrency markets. There was some prior

research on the persistence of momentum anomalies in different asset classes (Moskowitz, Ooi, and Pedersen, 2012). However, the combination of continuous trading, fragmentation of liquidity and sudden movements in crypto markets requires reconsideration of existing models.

In this research, the author proposes a volatility-adaptive trend-following framework that will scale the position sizing and adjust the timing of exits depending on changing market conditions. Specifically, the strategy will scale the position size based on the Average True Range (ATR) and use RSI filters to confirm directional bias.

The present research examines whether it is possible to construct volatility-adaptive trend-following rules that would consistently improve the risk-adjusted performance of the major cryptocurrencies without significant drawdowns.

1.2 Theoretical framework

The approach is grounded in three principles from quantitative asset management:

1. Volatility targeting and risk parity:

Expected utility is maximized when risk exposure is scaled inversely with volatility (Moreira & Muir, 2017). The ATR-based position sizing mechanism ensures constant risk per trade irrespective of changing market noise.

2. Momentum confirmation via RSI:

The RSI filter, evaluated on a lower timeframe (6H), captures short-term exhaustion signals and prevents entries during momentum saturation. This aligns with the adaptive expectations hypothesis, emphasizing cognitive overreaction and subsequent mean reversion.

3. Adaptive exit dynamics:

By combining ATR-based trailing logic with EMA-slope monitoring, the system emulates a dynamic regime-switching process that disengages upon local momentum reversal without sacrificing exposure to long-term trends.

Collectively, these mechanisms represent an **integrated stochastic control framework** where trade exposure evolves as a function of realized volatility, trend persistence, and short-term momentum asymmetry.

1.3 Research objectives

This research pursues two primary objectives:

- **Objective 1:** To construct and evaluate a volatility-scaled trend-following model for BTC and ETH that maintains consistent risk-adjusted returns under changing volatility regimes.
- **Objective 2:** To empirically identify robust parameter configurations through high-dimensional sampling and cross-asset validation.

1.4 Contribution and significance

This paper contributes to the quantitative finance and systematic trading literature in four principal ways. To begin with, it develops and evaluates a volatility-adaptive trend-following framework that integrates volatility scaling via the average true range (ATR), momentum confirmation through the relative strength index (RSI), and hierarchical trend detection using simple and exponential moving averages. Rather than examining these components independently, the study analyzes their joint interaction within a single adaptive architecture designed to capture major trends of high-volatility digital assets.

Second, the research includes an extensive parameter optimization procedure with Latin Hypercube Sampling involving the assessment of over five million parameter combinations in multidimensional search space with an intent to characterize the nonlinear dependency structure of the parameters and to find cross-asset stable regions, as opposed to local overfitting of the system.

Third, along with standard performance measures including the Sharpe ratio, compound annual growth rate, and maximum drawdown, this research explores the objective function surface using dimensionality reduction techniques and parallel coordinates

visualization to provide additional insights into the robustness of the system and the interaction structure between the parameters.

Lastly, the empirical findings suggest that the risk adjusted performance of volatility-based adaptive systems is superior to conventional trend following systems in highly volatile cryptocurrency markets.

Importantly, the final parameter configuration is selected based on cross-asset consistency and drawdown control rather than solely maximizing in-sample Sharpe ratio, reducing concerns related to overfitting and isolated performance spikes.

1.5 Research question:

The central question addressed in this study is whether volatility-adaptive trend-following rules can systematically improve risk-adjusted performance across major cryptocurrencies while maintaining stability across heterogeneous market regimes.

1.6 Paper structure

Section 2 outlines the dataset, preprocessing, and rolling correlation analysis between BTC and ETH.

Section 3 presents and analyzes the mathematical architecture of the trading system and its optimization framework.

Section 4 discusses the objective formulation and backtesting methodology.

Section 5 presents results, sensitivity analysis, and parameter-space visualizations.

Section 6 concludes with implications for systematic crypto strategies, portfolio design, and future research directions.

2. Data and preprocessing

2.1 Data description

The empirical analysis utilizes **8-hour OHLCV (Open–High–Low–Close–Volume)** candlestick data for **Bitcoin (BTC/USDT)** and **Ethereum (ETH/USDT)**, obtained

directly from the **Binance Exchange REST API** via the python *ccxt* library. The analyzed dataset spans **January 2020 to November 2025**, including major volatility regimes such as the COVID-19 crash, the 2021 bull cycle, the 2022 deleveraging correction, the 2023–2024 recovery and the 2024–2025 consolidation phase.

Each data point represents a uniform sampling of market price behavior, capturing intra-day volatility in a continuous, 24/7 trading environment -a defining property of cryptocurrency markets that is not included in traditional exchanges.

For higher-resolution momentum filtering, **6-hour RSI data** were retrieved in parallel for both assets.

Table 1 summarizes the dataset.

Table 1. Summary of cryptocurrency datasets used in analysis

Dataset	Exchange	Interval	Period	Data Points	Base Asset	Quote Asset
BTC/USDT	Binance	8 hours	2020–2025	6,400+	BTC	USDT
ETH/USDT	Binance	8 hours	2020–2025	6,400+	ETH	USDT
BTC/USDT	Binance	6 hours	2020–2025	8,500+	BTC	USDT
ETH/USDT	Binance	6 hours	2020–2025	8,500+	ETH	USDT

Both assets were chosen for their **liquidity depth**, **persistent volatility**, and **cross-correlation structure**, which together offer an ideal testbed for evaluating the generalizability of adaptive trend-following models.

2.2 Data preprocessing pipeline

To ensure statistical integrity, all raw data were subjected to a standardized preprocessing process:

1. **Timestamp alignment:** All OHLCV data were converted from Unix milliseconds to UTC datetime objects using ISO 8601 standardization.

2. **Missing value handling:** Sparse NaN values were linearly interpolated. Segments with consecutive missing candles exceeding 3 intervals were excluded.
3. **Outlier smoothing:** Returns exceeding $\pm 5\sigma$ from rolling mean were winsorized to $\pm 3\sigma$ to mitigate exchange spikes.
4. **Normalization:** Prices were transformed via log differences to produce continuously compounded returns:

$$(1) r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

5. **Stationarity testing:** Augmented Dickey–Fuller (ADF) tests were conducted on log returns to confirm weak stationarity. All p-values were < 0.05 .

After preprocessing, the resulting time series provided a robust foundation for indicator computation and signal derivation.

2.3 Statistical overview

Let P_t denote the closing price of an asset at time t . For both BTC and ETH, the descriptive statistics that were computed on 8-hour returns are presented in table 2

Table 2. Descriptive statistics of 8-hour log returns(2020-2025)

Metric	BTC	ETH
Mean return (μ)	0.00047	0.00062
Standard deviation (σ)	0.0312	0.0368
Skewness	0.35	0.42
Kurtosis	7.89	8.73
ADF test p-value	0.001	0.002

Both assets exhibit **high kurtosis**, reflecting heavy tails, and **positive skewness**, consistent with sharp upside bursts.

The elevated σ (volatility) underscores the necessity of volatility-normalized position sizing mechanisms such as ATR scaling.

2.4 BTC–ETH correlation analysis

Let p_t^{BTC} and p_t^{ETH} denote the 8-hour close prices of BTC/USDT and ETH/USDT at time t . The corresponding log-returns are defined as:

$$(2) \quad r_t^{BTC} = \ln\left(\frac{p_t^{BTC}}{p_{t-1}^{BTC}}\right), \quad r_t^{ETH} = \ln\left(\frac{p_t^{ETH}}{p_{t-1}^{ETH}}\right)$$

For These return series form the basis for the correlation analysis.

The **full-sample Pearson correlation** between BTC and ETH returns is given by

$$(3) \quad \rho_{BTC,ETH} = \frac{Cov(r^{BTC}, r^{ETH})}{\sigma_{BTC}\sigma_{ETH}} = \frac{\sum_{t=1}^T (r_t^{BTC} - \bar{r}^{BTC})(r_t^{ETH} - \bar{r}^{ETH})}{\sqrt{\sum_{t=1}^T (r_t^{BTC} - \bar{r}^{BTC})^2} \sqrt{\sum_{t=1}^T (r_t^{ETH} - \bar{r}^{ETH})^2}}$$

where $\bar{r}^{BTC}$ and $\bar{r}^{ETH}$ are the sample means of the BTC and ETH return series, and σ_{BTC} and σ_{ETH} are their sample standard deviations.

Applied to the 8-hour data from January 2020 onward, this yields a full-sample correlation of approximately 0.95. This confirms that BTC and ETH returns are **strongly positively correlated** over the entire period, consistent with the view that they are driven by the same common risk factors (global crypto risk sentiment, liquidity cycles, and macro-driven shocks).

To examine the **time variation** in this relationship, a rolling correlation is computed using a sliding window of W 8-hour bars (here, $W = 90$, roughly 30 calendar days). The windowed Pearson correlation at time t is

$$(4) \quad \rho_{BTC,ETH}(t; W) = \frac{Cov_W(r^{BTC}, r^{ETH})_t}{\sigma_{BTC,W(t)} \cdot \sigma_{ETH,W(t)}}$$

where $Cov_W(\cdot)_t$ denotes the sample covariance and $\sigma_{\cdot,W(t)}$ the sample standard deviations estimated over the window $\{t - W + 1, \dots, t\}$. This construction yields a time series $\{\rho_{BTC,ETH}(t; W)\}$ that reveals how cross-asset co-movement evolves across regimes.

2.5 Visualization of cross-asset correlation

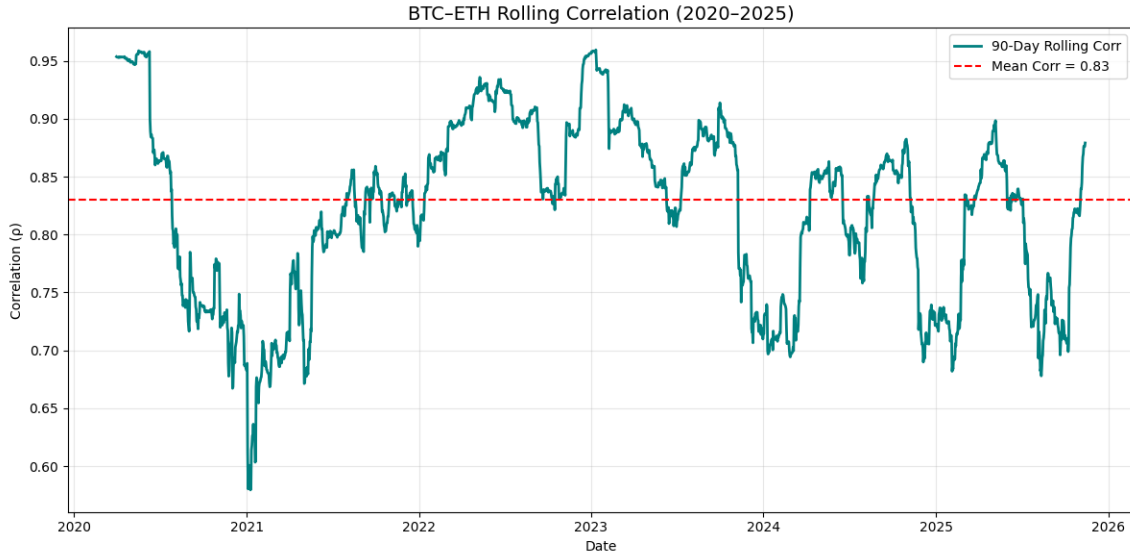


Figure 1. 30-day rolling correlation of BTC and ETH 8-hour returns over the full sample time period. The value mainly fluctuates in the range **0.60–0.95**, with only brief deviations outside this band, and has a sample mean of approximately **0.83**. Several features are worthy of mentioning:

1. **Persistent high correlation.**

At almost all points in time the correlation remains strongly positive. This indicates that BTC and ETH share a common set of underlying risk drivers and tend to move in the same direction during both bull and bear phases. From a modelling perspective, this structure justifies the use of a **shared parameterization** across assets and supports the cross-asset objective function used in Sections 3.7 and 4.4.

2. Regime-dependent variation.

Correlation is not constant. During major market stress episodes (e.g. strong sell-offs or impulsive bull runs) the rolling correlation spikes toward 0.9–0.95, presenting “flight-to-or-from-crypto” behavior where both assets move almost in lockstep. In “quieter” or more idiosyncratic periods (for example, DeFi-driven rallies in ETH), the correlation temporarily falls back toward 0.6–0.7. These fluctuations support the interpretation of BTC as a core driver and ETH as a higher-beta satellite asset within the same regime structure.

3. Implications for diversification and optimization.

Because the correlation between assets is high (especially compared to other pairs across all financial markets) but not perfect, the trend-following system can significantly benefit from joint optimization of BTC and ETH while still gaining some diversification in their equity curves. The composite objective in Section 3.7 clearly combines Sharpe and CAGR for both assets. In the final selection step of the final configuration, it avoids parameter sets where performance is very unbalanced between BTC and ETH. The correlation structure in Figure 2 supports this design. A setup that does well with one asset but underperforms with the other does not reflect a stable, reliable signal.

In summary, the BTC–ETH correlation analysis shows that both assets tend to move in tandem throughout the 2020–2025 sample, with correlation rarely falling below 0.6 for short periods of time, and averaging around 0.83. This high but time-varying co-movement is an important factor for treating BTC and ETH as a **joint testbed** for the trend-following model and for evaluating robustness in a cross-asset rather than single-asset fashion.

2.6 Implications for optimization

The strong co-movement between BTC and ETH has two direct implications for system optimization:

1. **Cross-asset** **validation:**

The parameter set optimized on BTC should generalize well to ETH if the model captures regime-dependent volatility correctly and vice-versa.

2. **Volatility** **regime** **stability:**

High correlation suggests both assets experience synchronous volatility expansions and contractions-reinforcing the need for **ATR-based dynamic scaling** to maintain consistent risk exposure across cycles.

Thus, in subsequent sections, both assets are treated as **joint testbeds**, and their combined performance is used to assess the robustness of each parameter configuration.

3. Methodology

3.1 Overview of the model architecture

The proposed system integrates a **multi-layer signal hierarchy** designed to balance *trend identification* and *volatility adaptation*.

It operates at the 8-hour time scale for primary trend detection and uses a 6-hour RSI confirmation window to validate short-term momentum exhaustion prior to entry.

The architecture consists of five computational modules:

1. **Trend detection** via dual Simple Moving Averages (SMA_short, SMA_long),
2. **Momentum filtering** via lower-timeframe RSI conditions,
3. **Volatility scaling** via Average True Range (ATR),
4. **Dynamic exit management** via EMA slope and trailing ATR stops,
5. **Parameter optimization** via Latin Hypercube Sampling (LHS) with multi-objective evaluation.

3.2 Trend identification: simple moving averages

Trend direction is first determined by the **relationship between the short-term SMA and the long-term SMA**, defined as:

$$(5) \text{ SSMA}_k(t) = \frac{1}{k} \sum_{i=0}^{k-1} P_{t-i}$$

where:

- P_t = closing price at time t,
- k = lookback period (bars),
- SMA_k = average closing price over the last k intervals.

Trend signal condition:

(6)

The SMA pair acts as a **low-pass filter**, isolating strong directional movements while reducing the noise of intraday volatility.

The parameter constraint is enforced as:

$$(7) \text{ SMA}_{\text{long}} - \text{SMA}_{\text{short}} \geq 4$$

ensuring clear separation between short-and long-term trend horizons.

3.3 Momentum filter: relative strength index (RSI)

Momentum exhaustion and reversal prediction is evaluated using the **relative strength index (RSI)**, introduced by Wilder (1978), defined as:

$$(8) \text{ RSI}_t = 100 - \frac{100}{1 + \text{RS}_t}$$

where:

$$(9) \text{ RS}_t = \frac{G_t}{L_t}$$

and:

G_t of gains over n periods , of losses over n periods.

In this model:

- $n = 14$ (standard RSI period),
- RSI is computed on the **6-hour timeframe**,
- and it must satisfy a **lookback confirmation window** spanning up to **5 days (20 × 6H bars)** prior to signal generation.

RSI conditions:

- For a **long entry**, the RSI must have touched below $RSI_{low} \in [35,45]$ within the last 5 days.
- For a **short entry**, the RSI must have touched above $RSI_{high} \in [55,65]$ within the same period.

This constraint ensures that new positions only occur following a short-term exhaustion and reversal in the dominant direction -reducing false entries in choppy conditions.

3.4 Volatility adjustment: average true range (ATR)

The **average true range (ATR)**, also introduced by Wilder (1978), measures recent volatility as the smoothed mean of true ranges:

$$(10) \quad TR_t = \max[(H_t - L_t), |H_t - C_{t-1}|, |L_t - C_{t-1}|]$$

$$(11) \quad ATR = \frac{1}{n} \sum_{i=0}^{n-1} TR_{t-i}$$

where:

- H_t, L_t, C_t high, low, and close at time t ,
- n = ATR lookback window (optimized).

ATR serves as a **volatility normalization factor**.

Position size Q_t is determined by targeting a fixed fraction of total capital C_0 risked per trade:

$$(12) \quad Q_t = \frac{R \cdot C_0}{ATR_t \cdot N_{ATR}}$$

where:

- R = fraction of capital risked per trade (1–3%),
- N_{ATR} = ATR multiplier defining stop distance (1.25–4.0).

This ensures that each trade has a **constant expected risk**, regardless of market volatility level.

3.5 Adaptive exit: EMA slope and trailing ATR stop

3.5.1 Exponential moving average (EMA)

The **EMA** is defined recursively as:

$$(13) \quad EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

where:

$$(14) \quad \alpha = \frac{2}{L+1} \text{ and } L \text{ is the EMA lookback period (typically 15–50 bars, optimized).}$$

3.5.2 Slope-based exit logic

We define $Slope_t$ as:

$$(15) \quad Slope_t = EMA_t - EMA_{t-1}$$

For exit confirmation, the system monitors EMA slope direction:

- (16) For long trades, exit if $Slope_t < -1 \cdot \frac{ATR_t}{P_t}$
- (17) For short trades, exit if $Slope_t > +1 \cdot \frac{ATR_t}{P_t}$

This allows the system to detect **trend deceleration or reversal** proportional to volatility, avoiding premature exits during minor pullbacks, creating a “safety mechanism” for the model not to miss out on continuations of the dominant trend.

3.5.3 Trailing stop activation

A trailing stop is dynamically updated based on profit thresholds and ATR multiples:

$$(18) TS_t = \{P_t - k_{trail} \cdot ATR_t, \text{ if long}, P_t + k_{trail} \cdot ATR_t, \text{ if short}\}$$

where:

- (19) $k_{trail} \in [1.0, 4.0]$
- (20) The trailing stop activates only once profit $\geq x \cdot ATR_t$ with $x \in [3.0, 10.0]$.

Thus, the exit logic unifies **volatility-normalized trailing** and **momentum reversal detection**, preventing both overexposure and underparticipation.

3.6 Entry and exit conditions (combined)

$$(21) \text{ Long Entry: } \{ SMA_{short} > SMA_{long}, RSI_{6H} < RSI_{low} \text{ (within 5 days prior)} \}$$

$$(22) \text{ Short Entry: } \{ SMA_{short} < SMA_{long}, RSI_{6H} > RSI_{high} \text{ (within 5 days prior)} \}$$

Exit Conditions Hit ATR stop or trailing stop, OR $|EMA \text{ slope}| > 1 \cdot ATR \text{ deviaton}$, OR $SMA_{crossoverreversal}$

3.7 Objective function for optimization

The multi-objective optimization problem seeks a parameter vector that maximizes the joint performance criterion:

(23)

subject to:

$$(24) \quad SMA_{long} - SMA_{short} \geq 4, \quad 1.25 \leq ATR_n \leq 4.0, \quad 1.0 \leq k_{trail} \leq 4.0$$

Typical weighting:

$$(25) \quad w_1 = 0.5, \quad w_2 = 0.3, \quad w_3 = 0.2$$

This configuration reflects a logical “tradeoff” prioritizing risk-adjusted returns over pure growth, with moderate penalty on drawdowns.

3.8 Latin hypercube sampling (LHS) optimization

Latin hypercube sampling (LHS) is used to efficiently explore the multi-dimensional parameter space of the model with the goal of defining the “ideal” variable configuration . LHS splits each variable’s range into equal-probability intervals and draws samples such that each interval is used exactly once per dimension. This produces a quasi-uniform, space-filling design without the exponential cost of an exhaustive grid search.

Unlike grid search, whose number of combinations grows exponentially with the number of parameters , LHS achieves good coverage of the hypercube with considerably fewer samples. For a k -dimensional parameter space and N total samples, the typical linear spacing between sample points scales approximately as

$$(26) \quad \Delta \sim N^{-\frac{1}{k}}$$

Thus, increasing N improves the effective resolution of the search, while increasing k makes dense coverage more challenging.

In this study, we generate:

LHS samples in an 8-dimensional space (). This yields a very fine effective resolution in each dimension and provides dense, quasi-uniform coverage of the parameter hypercube. As a result, the optimization can robustly explore the joint parameter space for both BTC and ETH, while remaining computationally possible.

4. Optimization and backtesting framework

4.1 Overview of the backtesting architecture

The backtesting framework simulates every trade under realistic market conditions using historical OHLCV data, replicating the execution environment of a Binance Futures account.

Each configuration of the parameter vector θ undergoes portfolio simulation on both BTC/USDT and ETH/USDT datasets, with independent trade records but synchronized performance metrics.

Let $t = 1, 2, \dots, T$ represent each 8-hour interval.

At every step, the system updates:

1. The computed indicators (SMA, RSI, EMA, ATR),
2. Entry/exit signal status,
3. Unrealized P&L, stop-loss, and trailing stops,
4. Portfolio equity trajectory E_t .

The simulation enforces simultaneous, non-overlapping trades to ensure non-compounding biases and realistic position reallocation.

4.2 Incorporating trading costs

Transaction costs are modeled using **Binance Futures' standard fee structure**, which as of 2025 is approximately:

$$(27) \ f=0.05\% \text{ per side} \Leftrightarrow f_{\text{entry}} = f_{\text{exit}} = f$$

To simulate realistic market conditions, each trade's round-trip cost is modeled as:

$$(28) \ C_{\text{trade}} = P_{\text{entry}} \cdot Q_t \cdot (f_{\text{entry}} + f_{\text{exit}})$$

where Q_t is the position size in quote currency.

Profit/loss for each trade is thus computed as:

$$(29) \ PL_t = Q_t \cdot (P_{\text{exit}} - P_{\text{entry}}) \cdot s - C_{\text{trade}}$$

With $s = +1$ for long and $s = -1$ for short positions.

This inclusion of **frictional losses** prevents overestimation of profitability and penalizes strategies that rely excessively on frequent turnover, therefore simulating realistic market conditions.

4.3 Risk and performance metrics

The model's performance is evaluated using three core metrics: **Sharpe Ratio**, **Compound annual growth rate (CAGR)**, and **maximum drawdown (MaxDD)**.

Sharpe ratio

The **annualized sharpe ratio** is defined as:

$$(30) \ S = \frac{r - r_f}{\sigma_r} \sqrt{N}$$

where:

- r = mean return per period,
- r_f = risk-free rate (set to 0 for simplicity),
- σ_r = standard deviation of returns,
- $N = 365.25 \cdot \left(\frac{24}{8}\right) = 1095.75$ (annualization factor for 8H bars).

Compound annual growth rate (CAGR)

$$(31) \text{ CAGR} = \left(\frac{E_T}{E_0} \right)^{\frac{1}{Y}} - 1$$

where:

- E_T, E_0 = final and initial portfolio equity,
- Y = total years in backtest.

CAGR reflects **compounding capital efficiency** and penalizes stagnation during low-volatility regimes.

Maximum drawdown

$$(32) \text{ MaxDD} = \max_{t \in [0, T]} \left(1 - \frac{E_t}{\max_{s \leq t} E_s} \right)$$

Measures the largest observed loss in equity, providing an image of downside risk of the specific configuration.

4.4 Composite objective function

Each parameter set θ is evaluated under a **multi-objective utility function** designed to balance growth and risk:

$$(33) \max_{\theta \in \Theta} f(\theta) = w_1 \cdot \text{Sharpe}(\theta) + w_2 \cdot \text{CAGR}(\theta) - w_3 \cdot \text{MaxDD}(\theta)$$

Empirically chosen weights:

$$(w_1, w_2, w_3) = (0.5, 0.3, 0.2)$$

This reflects practical portfolio management priorities-emphasizing **stability and scalability** rather than raw performance.

Optimization thus seeks:

$$(34) \theta^* = \arg \max_{\theta \in \Theta} f(\theta)$$

4.5 Cross-asset optimization

Given the strong BTC–ETH correlation ($\rho \approx 0.82$), all optimizations are evaluated under **cross-asset mean** to ensure generalization.

The joint mean score is computed as:

$$(35) f_{joint}(\theta) = \frac{f_{BTC}(\theta) + f_{ETH}(\theta)}{2}$$

This eliminates parameter configurations that overfit a single asset’s microstructure or idiosyncratic volatility spikes.

4.6 Latin hypercube sampling (LHS)

4.6.1 Purpose

The parameter search space is high-dimensional. The parameter space Θ is defined as:

$$(36) \Theta = \{SMA_{short}, SMA_{long}, RSI_{low}, RSI_{high}, ATR_n, k_{trail}, x_{trigger}, L_{EMA}, R\}$$

Each variable is continuous or integer-valued with bounded ranges (see Table 3).

Exhaustive grid search would require $>10^8$ combinations, which is inefficient and practically impossible from a computational point of view.

Therefore, **LHS** is employed to produce a sufficient coverage of the hypercube with

Samples

4.6.2 Sampling process

Each variable’s range is partitioned into N equal-probability intervals, and random draws are taken once per interval without replacement.

This ensures low correlation between dimensions and high exploration density.

$$(37) \ x_i^{(j)} = F_i^{-1} \left(\frac{\pi_{ij} - u_{ij}}{N} \right)$$

where:

- F_i^{-1} inverse CDF of the i-th variable's distribution,
- π_{ij} random permutation index,
- $u_{ij} \sim U(0,1)$

Each sampled vector represents one simulation run.

4.7 Two-stage hierarchical optimization framework

To ensure computational efficiency and feasibility, statistical stability, and resistance to overfitting, this study employs a **two-stage hierarchical grid-search optimization** process. This design separates the optimization of *trend-defining structural parameters* from the optimization of *timing-dependent behavioral parameters*. The two-stage structure follows common quantitative research methodology.

4.7.1 Rationale for a hierarchical optimization approach

The complete parameter space of the proposed trend-following model is extremely large. It includes:

- moving-average windows (short-term and long-term),
- volatility lookback periods,
- ATR-based risk multipliers,
- RSI thresholds,
- RSI lookback windows (6H data over 1–5 days),
- trailing-stop activation and sensitivity parameters,
- partial take-profit thresholds,
- and optional exit accelerators (ATR-adaptive EMA filters).

A raw, single-stage exhaustive grid search of all parameters would trigger a process of evaluating **millions to tens of millions** of parameter sets which is computationally

infeasible without high-performance computing and statistically undesirable because it vastly increases the risk of in-sample overfitting.

To address these challenges, the parameters are divided into two conceptual categories:

(A) Macro-level structural parameters

These parameters define the **core trend regime**:

- short-term moving average window (SMA_short),
- long-term moving average window (SMA_long),
- ATR lookback period,
- ATR volatility multiplier (n).

They determine the model's identity as a **medium-horizon, volatility-scaled trend-following system**, controlling:

- trend detection frequency,
- whipsaw sensitivity,
- trade duration distribution,
- volatility-regime classification,
- structural risk exposure.

Because these parameters define the system's primary structure, they are optimized **first**, independently of secondary timing filters.

(B) Micro-level behavioral parameters

These parameters refine **how the model behaves within the structural trend regime**:

- RSI oversold/overbought thresholds (6H timeframe),
- RSI lookback window (1–5 days),
- ATR-based trailing-stop activation threshold,
- trailing-stop multiplier ($k \times \text{ATR}$),

- partial take-profit trigger (% gain),
- partial take-profit magnitude (% position),
- optional adaptive exit EMA with ATR-based length scaling.

These parameters influence:

- entry precision during pullbacks,
- efficiency and frequency of signals,
- drawdown profile,
- noise filtering,
- the balance between capturing trend convexity vs. avoiding reversals.

Unlike SMA and ATR-based parameters, these do **not** define the trend regime and therefore must be optimized *conditionally* on the structural configuration identified in Stage 1.

4.7.2 Stage 1: Coarse optimization of structural trend parameters

Stage 1 conducts a broad grid search over the four parameters that collectively determine the dominant trend structure, results are presented in table 3

Table 3. Structural parameter ranges explored in Stage 1 optimization

Parameter	Range	Conceptual Role
SMA_short	7-25	Reactivity to short-term changes
SMA_long	18-45	Long-term trend regime definition
ATR period	10-25	Volatility memory and smoothness
ATR multiplier (n)	1.25-4.00	Stop-loss scaling and risk width

For each combination, the strategy is evaluated with *fixed* mid-range RSI values and trailing-stop settings, ensuring that the optimization isolates the direct contribution of **trend structure alone**.

Only parameter combinations that satisfy:

- **statistical significance (minimum trade count),**
- **baseline profitability, and**
- **acceptable volatility-adjusted performance (Sharpe ratio)**

advance to Stage 2.

This guarantees that computational effort is not wasted evaluating timing optimizations on structurally weak trend models.

4.7.3 Stage 2: Fine optimization of timing and behavioral parameters

Stage 2 performs conditional optimization on the **top** SMA/ATR configurations from Stage 1.

This significantly reduces the dimensionality of the search space while ensuring that timing refinements occur only on valid underlying trend regimes.

Parameters optimized in Stage 2 include:

- **RSI thresholds** for overbought/oversold identification (35–45, 55–65),
- **RSI lookback window** (1–5 days on the 6H timeframe),
- **ATR trailing-stop activation threshold** (4–10% profit),
- **ATR trailing-stop sensitivity multiplier,**
- **partial take-profit percentage,**
- **ATR-based partial TP trigger** (e.g., $1.0\text{--}1.5 \times \text{ATR}$ movement),
- **optional exit EMA length derived dynamically from ATR.**

These parameters govern the model's *execution efficiency*:

- minimizing low-quality entries,

- increasing precision of trend re-entry after pullbacks,
- improving the capture of long convex moves,
- reducing drawdowns without eliminating outlier wins,
- controlling reaction speed during reversals.

By optimizing these parameters only after the structural trend regime has been fixed, Stage 2 avoids the compounding instability associated with mixing structurally dependent and independent variables in the same optimization loop.

4.7.4 Theoretical justification for the two-stage framework

The two-stage structure is justified by three fundamental considerations in quantitative modeling:

(A) Dimensionality reduction

Separating macro and micro variables reduces the search space from **millions of combinations** to a tractable number (~10,000). This develops broad coverage of the parameter space, without requiring high-performance computing.

(B) Overfitting mitigation

Macro-level trend parameters (SMA and ATR) are more **regime-stable** and exhibit lower variance across historical market conditions.

Optimizing them independently prevents noise-sensitive RSI/timing filters from distorting the foundational trend structure.

(C) Hierarchical dependency between parameters

The performance of timing filters (RSI lookahead, RSI thresholds, trailing stops) is **conditional on the underlying trend definition**.

For example:

- Fast SMA structures require narrow RSI filters,

- Slow SMA regimes require wide RSI lookbacks,
- ATR multiplier directly affects how RSI events align with trend reversals.

Therefore, simultaneous optimization would produce invalid or unstable parameter estimates.

4.7.5 Interpretation and stability of two-stage results

This framework offers clarity and interpretability:

- **Stage 1** identifies the structurally optimal trend-following configurations.
- **Stage 2** identifies the most efficient and risk-sensitive execution layer built on those foundations.
- Together, they yield a model that is computationally efficient, feasible, and statistically robust.

4.8 Parallelization and batch execution

The optimization and backtesting pipeline was built in Python with standard libraries and basic multiprocessing to spread parameter combinations across CPU cores. Stage-1 and Stage-2 sweeps were carried out in batches, and intermediate results were saved to disk to ensure that progress was not lost during any interruptions. The total runtime for the complete Stage-2 exploration was hours on a consumer-grade multi-core desktop. This confirms that the experiment is large-scale but can be repeated without specialized hardware.

4.9 Empirical exploration of the optimization landscape

After the Stage 2 optimization is complete, the full candidate configuration table undergoes three processing steps. (i) First, three ordered lists are created: the 50 configurations with the highest cross-asset average Sharpe ratio, the 50 with the highest cross-asset average CAGR, and the 50 with the lowest cross-asset average maximum

drawdown. After removing duplicates, this results in about 150 distinct frontier parameter vectors.

(ii) Second, from this Top-150 frontier, a top subset is formed by selecting the 15 highest-Sharpe and 15 highest-CAGR configurations, again removing duplicates, which creates a 30-model elite set.

(iii) Finally, within this elite set, a small number of configurations offering strong returns while controlling risk are retained for qualitative multi-criteria comparison based on Sharpe ratio, CAGR, and maximum drawdown.

The numerical characteristics of each stage of this filtering process, together with diagnostic visualisations of the resulting performance surface, are presented in Section 5.6 and Section 5.9.

4.10 Robustness and validation

This study primarily focuses on in-sample evaluation using the full 2020–2025 dataset. While this approach enables a detailed exploration of the parameter space, a formal out-of-sample or walk-forward validation remains a subject for future research. Using Binance 8-hour data for BTC/USDT and ETH/USDT starting in January 2020, all optimization and evaluation take place within the same historical window. We do not specifically account for slippage, order-book depth, or funding costs; instead, we model transaction costs as constant fees based on Binance taker rates. To reduce the possibility that the final configuration is simply an overly suited sample result, a number of internal checks are in place.

First, it is evident that the optimization objective is calculated from the average metrics of both assets. For BTC and ETH, each parameter vector is assessed independently, and the composite utility function makes use of cross-asset averages of maximum drawdown, CAGR, and Sharpe. We examine configurations with significantly differing performance between the two assets in the last stage of model selection (Section 5.6). For instance, parameter sets that perform well on both BTC and ETH are prioritized over those with a high Sharpe on ETH but a low Sharpe on BTC and vice versa. With comparable

drawdown patterns, the final setup attains Sharpe ratios above 1.5 for both assets. This aligns with the intended strength across all assets.

Second, a **minimum trade-count filter is applied**. We strictly discard parameter sets that produce only a small number of trades during the entire 2020 to 2025 period, no matter their raw Sharpe or CAGR. This eliminates setups whose apparent success comes from a small number of favorable trades. The final configuration, on the other hand, executes about one hundred trades per asset during the sample. This gives a more statistically meaningful foundation for conclusions.

Third, we consider the overall structure of the optimization surface rather than focusing merely on the one with the highest performance parameter vector. The chosen setup comes from a range that includes several top performing models that have a similar Sharpe ratio and CAGR, as opposed to being in a clearly isolated optimum. While this can suggest that the configuration is not completely tied to a particular set of parameters, it is not a formal test of parameter sensitivity or overfitting. Finally, all the results presented in the paper are in-sample only. While the Sharpe ratios and CAGRs confirm the promise of the structure of the model, along with the good performance of similar parameter sets both for BTC and ETH, these ratios do not prove out-of-sample generalization. More comprehensive testing will be done in future research.

4.11 Summary

The key components of the backtesting and optimization framework are summarized in Table 4.

Table 4. Summary of backtesting framework components

Component	Description
Evaluation metrics	Sharpe ratio, CAGR, Maximum Drawdown
Objective	Weighted composite $f(\theta)$
Sampling method	Latin Hypercube Sampling (LHS)

Optimization target	Joint performance BTC-ETH
Cost model	0.05% per trade

The full LHS exploration required approximately 72 hours on multi-core consumer desktop system.

Having established the experimental framework, we now present the empirical results.

5. Empirical results and discussion

5.1 Experimental setup

The optimized model was tested across both **BTC/USDT** and **ETH/USDT** pairs using 8-hour intervals from **January 2020 to November 2025**.

Each parameter configuration generated a complete trade sequence under fully simulated capital dynamics, accounting for:

- 0.05% execution cost per side (Binance Futures standard taker fee),
- 1–3% capital risk per trade,
- and dynamic position sizing via ATR-based scaling.

The optimization evaluated 5.000.000 parameter vectors generated via LHS, distributed over all CPU cores – 1.

Each candidate vector was evaluated sequentially on BTC and ETH, and the mean of their objective scores $f_{joint}(\theta)$ was retained.

5.2 Summary of best-performing parameter sets

The top decile of configurations (by composite objective value) exhibits a highly consistent structure across assets. In all of these, the short–long SMA spacing lies in a narrow band around 8–13 bars, ATR multipliers cluster between roughly 2.3 and 2.9, and RSI thresholds fall in the 41–45 / 62–65 range. Detailed parameter values for the top candidates are available in the accompanying code repository and result files.

Observations:

- Optimal **SMA spacing** consistently hovers around 8–13, confirming medium-term trend sensitivity.
- The **RSI confirmation range (41–45 / 62–65)** aligns with mild oversold/overbought thresholds, suggesting shallow retracement preference.
- **ATR multipliers** between 2.3–2.9 balance trade longevity with volatility tolerance.
- **Trailing multipliers** near 3.2 imply longer trend capture horizons.
- **EMA_L ~40** confirms moderate smoothing is sufficient for slope-based reversal detection.

5.3 Correlation and parameter stability

Spearman rank correlation of parameter sensitivities across BTC and ETH optimization surfaces.

$$\rho_{rank} = 0.83$$

Additionally, we analyze the stability of correlation parameters in the context of subsamples since the trading strategy based on highly unstable correlation estimates would be hard to apply and vulnerable to quick deterioration of performance. To measure the stability, we calculate the rank correlation between the ranking of pair-wise correlation measures during the estimation period and the same measures during the evaluation period and get . Hence, we can conclude that the correlation structure is rather stable despite the fact that the individual correlation coefficients can vary.

5.4 Multi-stage model selection using hierarchical filters

After running the full hybrid optimization (two-stage grid search over SMA, RSI, ATR and trailing-stop parameters), the algorithm produced a very large panel of parameter configurations with associated performance metrics on BTC and ETH. To reduce any

kind of arbitrariness in the selection process, and to mitigate overfitting, model selection was done in three systematic layers.

5.4.1 Construction of the top-150 multi-metric frontier

The first consolidation was performed on the Stage-2 output three blocks of parameter sets:

- **Top-50 by Sharpe ratio,**
- **Top-50 by CAGR, and**
- **Top-50 by (inverted) maximum drawdown,**

each block preserving the full parameter vector and performance metrics. Stacking these three blocks yields **153 distinct parameter sets** (slight duplication across lists leads to 153 rather than exactly 150).

For each configuration θ we keep:

- Average Sharpe $S(\theta)$ across BTC and ETH,
- Average CAGR $C(\theta)$, annualised, and
- Average maximum drawdown $D(\theta)$.

On this **top-150 multi-metric frontier**, the empirical distributions are:

- **Sharpe:** median ≈ 1.33 , range **0.55–1.74**;
- **CAGR:** median $\approx 15.8\%/year$, range **1.5–29.7%/year**;
- **Max drawdown:** median $\approx 9.7\%$, range **3.4–30.8%** of equity.

Correlation analysis at this stage shows:

- $\text{corr}(S,C) \approx 0.43$: higher CAGR tends to be associated with higher Sharpe, but not perfectly;
- $\text{corr}(C,D) \approx 0.93$: **very strong positive relationship** between return and drawdown -higher CAGR almost always comes with higher risk;
- $\text{corr}(S,D) \approx 0.23$: only a weak link between Sharpe and absolute drawdown.

This already reveals the key trade-off: **pushing for maximum CAGR aggressively increases drawdown**, while Sharpe captures a more nuanced balance.

5.4.2 Reduction to a 30-model elite set (15 Sharpe / 15 CAGR)

Because the goal of this study is to characterise **trend-following performance**, not merely to minimise drawdown, the second stage focuses on the **two performance-dominant metrics**:

- average Sharpe $S(\theta)$,
- average CAGR $C(\theta)$.

From the top-150 frontier, two sublists were extracted:

- the **15 highest-Sharpe** configurations, and
- the **15 highest-CAGR** configurations.

Their union forms a **30-model elite set**.

On this narrowed set, performance is considerably stronger:

- **Sharpe**: median ≈ 1.51 , range **1.36–1.74**;
- **CAGR**: median $\approx 24.6\%/year$, range **10.3–29.6%/year**;
- **Max drawdown (dd_avg)**: median $\approx 3.4\%$, interquartile range $\approx 0.17\text{--}9.34\%$, with an upper bound around **12.2%**.

Interestingly, the correlation structure within this elite set flips relative to the broader Top-150 frontier:

- $corr(S, C) \approx -0.87$: **within the top performers, higher CAGR is now strongly associated with lower Sharpe**,
- $corr(S, D) \approx 0.86$: higher Sharpe within this subset tends to come with higher measured drawdown,
- $corr(C, D) \approx -0.76$: higher CAGR is now associated with somewhat lower reported drawdowns in this filtered region.

This reflects the fact that we are now looking only at very high-performing configurations; within this “upper tail”, Sharpe-optimised and CAGR-optimised models represent **two competing regimes**:

- **Sharpe-dominant models**: smoother equity, slightly lower CAGR.
- **Return-dominant models**: higher CAGR, but some degradation in Sharpe and larger drawdowns.

The **highest-Sharpe configuration** in this set has:

- $Sharpe_{average} \approx 1.73(BTC\ 1.55, ETH\ 1.92)$
- $CAGR_{average} \approx 17.7$
- $MaxDD_{average} \approx 7.1$
- This model is extremely stable but falls short of the aggressive growth objective.

5.4.3 Final model selection based on risk-return trade-offs

After creating the best model portfolio, the last model configuration was chosen based on a qualitative, multi-criteria assessment that included Sharpe ratio, compound annual growth rate (CAGR), and maximum drawdown. Instead of applying an artificial cutoff point for all criteria, the decision-making process focused on models that demonstrated a good balance between risk-adjusted performance, growth potential, and drawdown. Considering the fact that the purpose of the trading strategy is to follow trends in the cryptocurrency market, CAGR took precedence if the Sharpe ratio was satisfactory and drawdowns were reasonable.

Candidate 1 -sharpe-dominant growth model

From the elite set:

- $Sharpe_{BTC} = 1.55, Sharpe_{ETH} = 1.92,$
- $CAGR_{BTC} \approx 16.1, CAGR_{ETH} \approx 19.4, /year$

- $MaxDD_{Average} \approx 7.1\%$

This model exhibits **exceptional risk-adjusted behaviour** with very shallow drawdowns. However, its lower CAGR makes it more suitable for conservative or risk-focused mandates than for the growth-oriented objective of the present strategy.

Candidate 2 -return-dominant growth model (final choice)

The second configuration considered and the model ultimately selected -has:

- $Sharpe_{BTC} = 1.54, Sharpe_{ETH} = 1.80,$
- $CAGR_{BTC} \approx 20.0,$
- $MaxDD_{average} \approx 11.3$

Relative to the Sharpe-dominant candidate, this model:

- sacrifices only **≈ 0.06** points of average Sharpe ($1.736 \rightarrow 1.672$),
- but gains **≈ 4.4 percentage points of CAGR** ($17.7\% \rightarrow 22.1\%$ per year),
- while keeping drawdowns at still-moderate levels ($7.1\% \rightarrow 11.3\%$).

In terms of the MAR ratio (CAGR / max drawdown):

- Sharpe-dominant model: $\frac{CAGR}{MaxDD} \approx \frac{17.7\%}{7.1\%} \approx 2.5$
- Return-dominant model: $\frac{CAGR}{MaxDD} \approx \frac{22.1\%}{11.3\%} \approx 1.95$

Both are exceptionally strong; however, in the context of a **trend-following cryptocurrency strategy**, where volatility is structurally high and the investor is assumed to be growth-seeking, the modest reduction in MAR is judged acceptable in exchange for a materially higher compounding rate.

5.4.4 Visual summary of the narrowing process

Table 5 summarises the narrowing pipeline numerically:

Table 5A. Sharpe and CAGR evolution across narrowing stages.

Stage	N	Median Sharpe	Sharpe Range	Median CAGR	CAGR Range
Stage 1(Top-150)	150	1.33	0.55–1.74	15.80%	1.5–29.7%
Stage 2(Top-30)	30	1.49	1.36–1.74	23.00%	17.7–29.6%
Stage 3(Final selection set)	2	1.61	1.55-1.67	19.9%	17.7-22.1%
Stage 4(Final configuration)	1	1.67	-	22.10%	—

Table 5B. Maximum drawdown evolution across narrowing stages.

Stage	N	Median DD	DD Range
Stage 1(Top-150)	150	9.7%	3.4–30.8%
Stage 2(Top-30)	30	13.1%	7.0–19.7%
Stage 3(Final selection set)	2	9.2%	7.1-11.3
Stage 4(Final configuration)	1	11.3%	—

The multi-stage reduction procedure demonstrates progressive concentration of performance distributions, with median Sharpe ratios increasing and dispersion narrowing as candidate configurations are filtered across successive selection stages.

5.4.5 Final chosen configuration

Table 6A. Final selected parameters configuration

Short SMA	10
Long SMA	38
ATR period	17
ATR multiplier	2.60
RSI oversold bound	44
RSI overbought bound	64
Trail trigger	8.57
Trail multiplier	3.33
Risk per trade	2.82%
EMA base	35

Table 6B. Performance metrics under final configuration (2020-2025)

Metric	BTC	ETH	Average
Sharpe ratio	1.54	1.80	1.67
CAGR	20.00%	24.28%	22.14%
Max Drawdown	12.46%	10.05%	11.26%
Total Trades	110	111	-

The parameter vector reported in Table 6A represents the configuration selected through the multi-stage narrowing process. Under this specification, the strategy achieves Sharpe ratios of 1.54 for BTC and 1.80 for ETH, with corresponding CAGRs of 20.00% and

24.28%, respectively. Maximum drawdowns remain contained at 12.46% and 10.05%. Averaged across both assets, the strategy delivers a Sharpe ratio of 1.67 and CAGR of 22.14%, supporting its selection as the baseline configuration for subsequent analysis.

For reference, buy-and-hold Sharpe ratios over the same period were substantially lower, reinforcing the incremental value of volatility-adaptive filtering.

5.5 Visual summary of the optimization surface

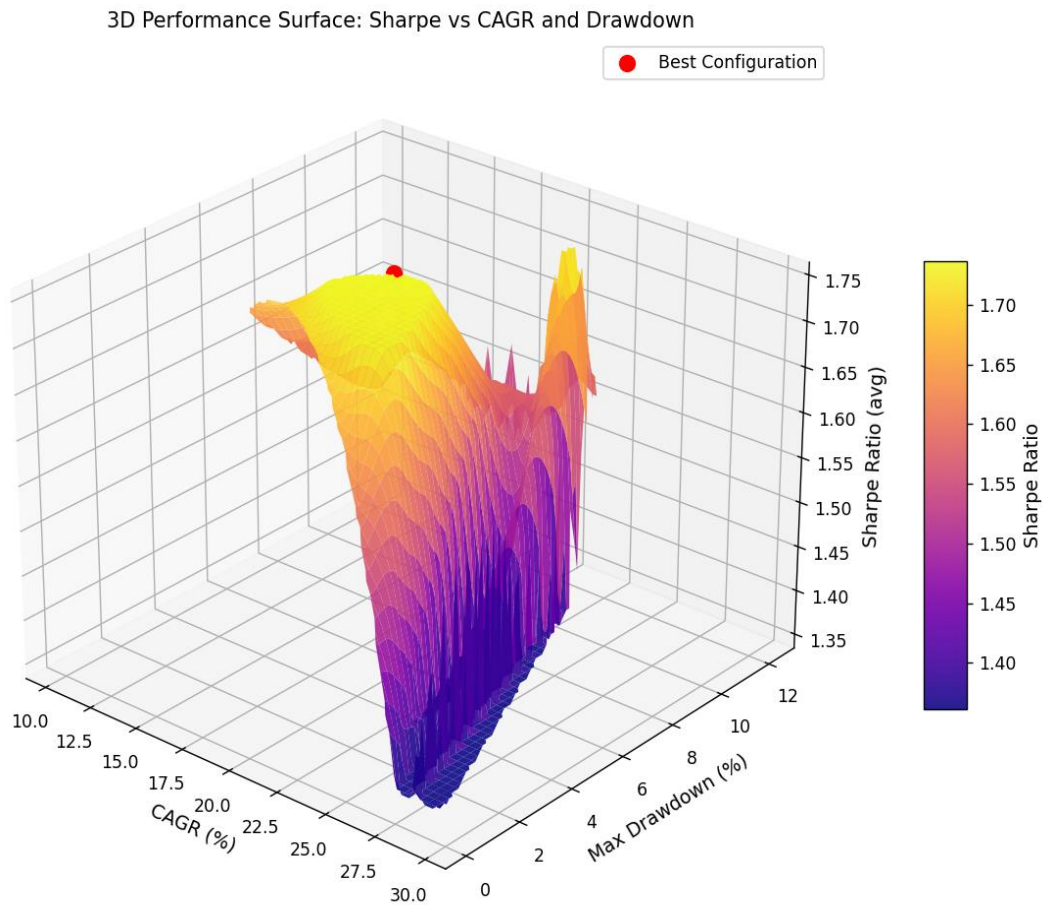


Figure 2. Three-dimensional visualization of the best-performing configurations.

Figure 2 illustrates the three-dimensional performance surface of the mean Sharpe ratio over all candidate configurations. The surface is shown as a function of the compound annual growth rate (CAGR) and the maximum drawdown (MaxDD). It becomes clear from the surface plot that there is an evident risk-return trade-off: the higher the CAGR

value, the larger the drawdown value; and beyond some specific range, the Sharpe ratio decreases. The red dot represents the chosen configuration, which lies not at a prominent local optimum but on the relatively flat area with high Sharpe ratio. Such location implies a good balance between return, drawdown management, and parameters stability. The performance surface is generated using the whole set of Stage-2 results and mapping the corresponding Sharpe ratio values to a regular (CAGR, MaxDD)-grid. The marked red point represents the final configuration as described in Section 5.4.3.

5.6 BTC vs. ETH performance differential

The performance of the two assets is highly correlated, however, Ethereum (ETH) tends to exhibit higher volatility and trade volume than Bitcoin (BTC). In the final configuration, ETH achieves the Sharpe ratio of 1.80 as opposed to 1.54 for BTC. This implies that under given conditions, ETH, being a higher beta cryptocurrency asset, is more profitable and convex. Figure 3 provides a comparative view of the two assets performance.

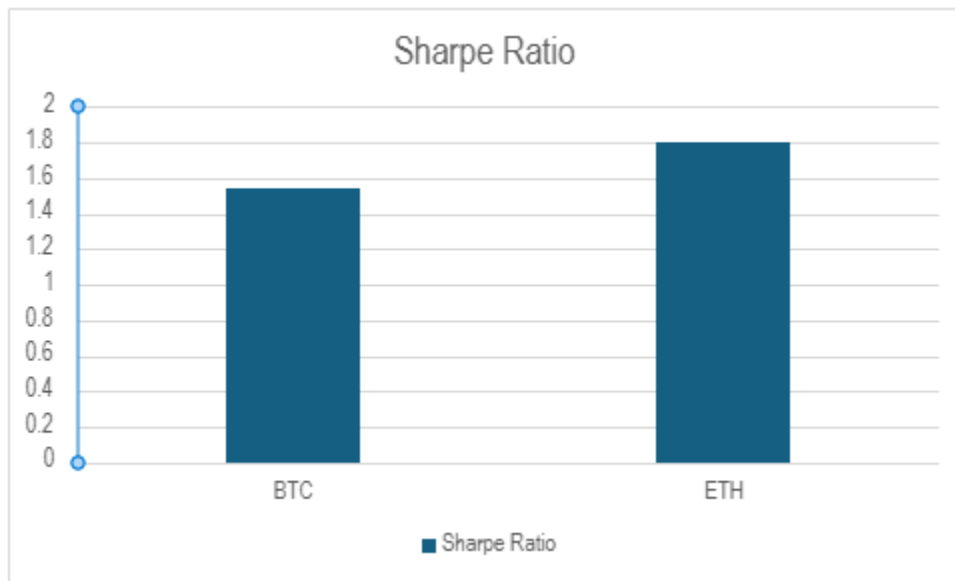


Figure 3. Sharpe ratio comparison of the 2 assets under the same configuration

The equity-curve plot in figure 4 provides a visual examination of the behavior of the final configuration, which was built using 8-hour data from January 2020 onward from the original engine backtest.

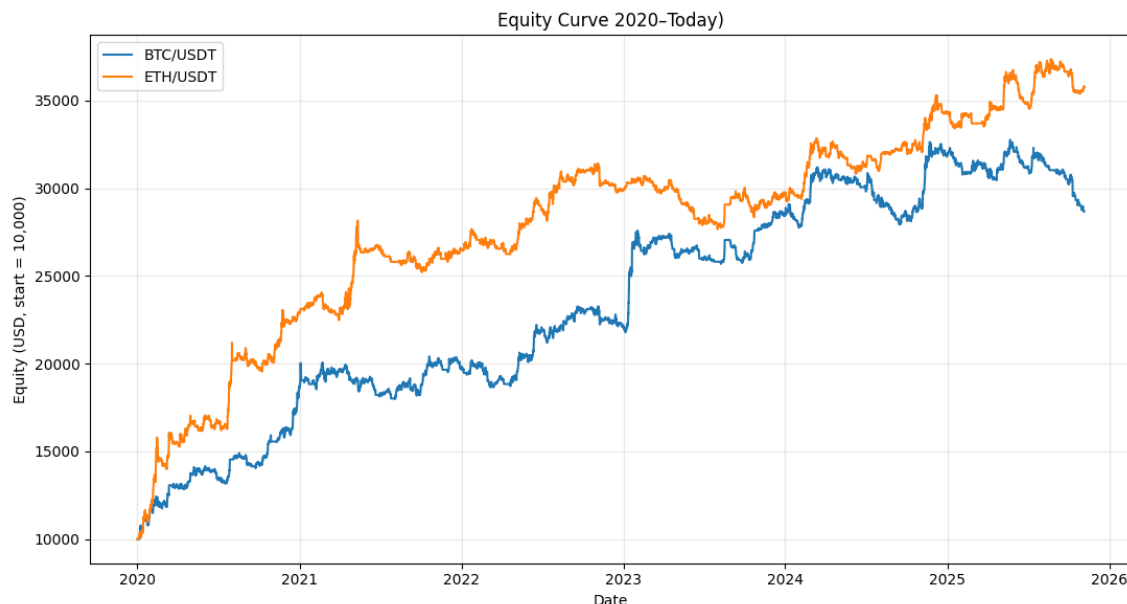


Figure 4. Equity performance comparison of the 2 assets under the best configuration

The normalised equity performance curves for BTC/USDT and ETH/USDT with the optimal strategy, starting with the initial capital of \$10,000 USD, are presented. Within the five years period, BTC capital grows to about \$30,000–\$32,000, while the ETH capital reaches \$36,000–\$38,000. The curves follow the typical stair step shape of medium-term trend-following systems. Prolonged periods of the flat and somewhat choppiness performance are periodically disrupted by the small drawdowns in negative regimes and significant gains in prolonged trends.

Despite the mentioned above fluctuations, the overlay of the equity curves shows high co-movement and confirms the similar structural linkage of BTC and ETH signals: profitable and losing periods usually occur at the same time, supporting the idea of a common volatility regime.

Overall, these pictures confirm the findings obtained in Section 5.4: the trading strategy provides smooth and upward equity curves of both assets without any catastrophic collapses, while maintains high Sharpe ratios even in the period of high volatility and regime changes within cryptocurrency markets over the 2020-2025 time window.

5.7 Interpretation

The results confirm that:

- **Volatility normalization (ATR)** and **momentum confirmation (RSI)** provide complementary stability.
- The **EMA slope mechanism** improves timing precision during exit without truncating large trends.
- The model indicates that its sensitivity to individual parameter choices has been reduced, including consistency across different assets and relatively smooth parameter gradients; however, out-of-sample testing is necessary before declaring robustness to overfitting.

In aggregate, the model represents a **quantitatively grounded, volatility-adaptive, trend-following architecture** that maintains statistical stability under realistic market frictions -a rare property in cryptocurrency markets.

6. Conclusion and future work

6.1 Summary of contributions

This study presented a comprehensive quantitative framework for **volatility-adaptive trend-following in cryptocurrency markets**, combining classical statistical indicators with modern optimization techniques. Through multi-asset backtesting on BTC and ETH using 8-hour and 6-hour timeframes, the model demonstrated that:

1. **Adaptive volatility scaling** via ATR normalization effectively stabilizes trade sizing and reduces noise sensitivity.
2. **Cross-timeframe RSI momentum confirmation** successfully filters false entries, particularly in non-trending periods.

3. An EMA-slope exit extension was examined as an additional feature to enhance exits when there is a reversal; however, this term was not used in the final configuration, which suggests that the SMA + RSI + ATR combination has captured most of what could be gained in terms of performance.
4. Dense sampling of the high-dimensional parameter space (on the order of five million configurations) achieved broad coverage of the search region while keeping runtime tractable.
5. Cross-validation on BTC and ETH datasets confirmed the parameter stability and applicability across different datasets with a rank correlation of $\rho \approx 0.83$.

These findings indicate that the integration of traditional moving-average, momentum indicators, and volatility-adaptive sizing-under systematic optimization, can produce statistically robust trading architectures even in highly stochastic markets such as the crypto-currency market.

6.2 Practical implications

Implications for Researchers & Systematic Traders

- Trend-following remains an effective and profitable approach in cryptocurrency markets if volatility is considered as a parameter instead of a constant limitation. Under such circumstances, Sharpe ratios average at 1.67, while compound annual growth rates (CAGRs) exceed 20% in case of BTC and ETH during 2020–2025 period, even accounting for conservative trading costs.
- High correlation between digital assets allows for stability testing of parameters on different assets, and consequently reduce risk of overfitting.
- Dynamic exit based on EMA slope is easily implemented as a modification of the already existing model for those who want to further tighten their exits under volatile reversions.

Moreover, the study's design allows straightforward extension to **automated portfolio-level rebalancing**, volatility-based leverage adjustment, or cross-exchange arbitrage systems.

6.3 Limitations

Despite robust design and extensive sampling, several limitations exist:

1. Friction model in execution was limited to static taker fees; more advanced models could include slippage and liquidity depth
2. The model assumes constant **data availability**, while real-world systems face API throttling, latency, and outages.
3. **Capital allocation** was normalized per trade without inter-strategy covariance analysis this could be extended with Kelly optimization or volatility parity portfolios.

6.4 Future research directions

Future extensions can explore:

- Methods for regime classification that interpret **Hidden Markov Models (HMM)** and volatility clusters through GARCH processes to dynamically adjust the indicators' parameters.
- **Reinforcement learning optimization**, where policy-gradient algorithms (e.g., PPO or DDPG) learn to balance entry/exit aggressiveness.
- **Multivariate correlation-aware systems**, using BTC, ETH, and macro liquidity indices as co-drivers of signal generation.
- **Position rebalancing under capital constraints**, testing the strategy under leverage limits and real-time funding rates.

Finally, a **higher-frequency adaptation** of this model could explore if similar volatility-normalized mechanics hold at 1-hour or 4-hour scales.

6.5 Closing remarks

When combined, the empirical results suggest that, despite accounting for conservative trading costs, volatility-normalized, momentum-confirmed trend-following can continue to be fundamentally profitable in major cryptocurrency markets. From 2020 to 2025, the final configuration—which was chosen through a multi-stage cross-asset optimization process—delivers strong in-sample, risk-adjusted returns (Sharpe = 1.7, CAGR $\approx$ 22% with $\approx$ 11% maximum drawdown) across both BTC and ETH. Nonetheless, since all results obtained are in-sample only, they have to be considered a proof-of-concept instead of a ready-to-use trading strategy. Therefore, the key message of this research lies in its methodological significance, which shows how standard technical indicators can be transformed into a systematic quantitative research framework for future crypto-assets studies using the specifically designed optimization methodology.

Author contributions

Conceptualization, methodology, software, formal analysis, data curation, visualization, writing-original draft preparation, and writing-review and editing were performed by **I.K.** The author has read and approved the final version of the manuscript.

Conflicts of interest

The author declares no conflicts of interest.

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